ME228 — Project Final Report

Data-Driven Modeling of Fatigue Strength in Low-Alloy Steels

Mixture-of-Experts Prediction, Hyperparameter Optimisation, Inverse Alloy Design, and Cost Analysis

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1 Problem Statement

Fatigue failure accounts for 80–90% of mechanical structural failures [1]. Physical rotating-bending fatigue testing is destructive, slow, and expensive. The goal of this project is to **predict the rotating bending fatigue strength** (in MPa) of low-alloy steels from their chemical composition and heat-treatment parameters alone, and to **invert the model** to recommend the cheapest viable alloy configuration for a given operating condition.

1.1 Dataset

The NIMS Fatigue Dataset [1] contains **437 experimental records** with **26 input features** and one target variable (fatigue strength, MPa). Features span:

- **Chemical composition** (12 variables): C, Si, Mn, P, S, Ni, Cr, Cu, Mo, and minor elements.
- **Heat-treatment parameters** (13 variables): normalising temperature (NT), through-hardening (THT, THt, THQCr), carburising (CT, Ct), diffusion (DT, Dt), quenching (QmT), tempering (TT, Tt, TCr).
- **Specimen geometry** (3 variables): reduction ratio (RedRatio), grain-size proxies (dA, dB, dC).

The target distribution is *bimodal*: carburised steels (CT = 930 °C, $N = 48$, 11%) reach fatigue strengths above 900 MPa, while the non-carburised majority ($N = 389$) clusters between 200 and 750 MPa.

2 Methodology

2.1 Mixture-of-Experts Split

A single global model fails on bimodal data because it averages across two physically distinct failure mechanisms. We engineer a binary flag:

$$\texttt{is_carburized} = \mathbf{1}[\,CT = 930\,]$$

and train **separate expert models** per metallurgical regime. This alone reduced RMSE by ∼30 % compared with our initial global baselines (Linear Regression, Ridge, Lasso).

2.2 Feature Selection Pipeline (VIF → SHAP)

With 26 raw features many processing variables are highly collinear (e.g. CT, DT, and QmT share thermal cycles). We apply a two-stage filter:

1. **Variance Inflation Factor (VIF)** — iteratively drop the feature with the highest VIF until all remaining features have VIF < 10. This removed 6 features per regime.
2. **SHAP importance scout** — fit a shallow Random Forest (150 trees, depth 5) on the VIF survivors. Retain the top-k features by mean |SHAP| across the training set ($k = 12$ for non-carburised, $k = 10$ for carburised).

The final feature sets are given in Table 1.

Table 1: Selected features per metallurgical regime after VIF→SHAP pipeline.

Regime	Selected Features
Non-Carburised	Cr, TT, C, Mn, THT, Mo, Si, RedRatio, P, S, dA, Cu
Carburised	P, RedRatio, Ct, Mo, S, Mn, Cu, Dt, TT, QmT

Key physical insight: For non-carburised steels, tempering temperature (TT) and chromium content (Cr) dominate — consistent with martensite stability theory. For carburised steels, carburising time (Ct) and reduction ratio (RedRatio) rank highest, reflecting case-depth and microstructural refinement effects.

2.3 Model Selection

- **Non-Carburised** ($N = 389$): XGBoost — handles the non-linear interaction between tempering temperature and alloy composition efficiently; large enough for ensemble depth.
- **Carburised** ($N = 48$): Random Forest — more robust to overfitting on small datasets (no boosting cascade); shallow depth and high min-samples constraints act as regularisers.

2.4 Hyperparameter Optimisation (Optuna + Nested CV)

We replace the single 80/20 split from Phase 1 with **5-fold cross-validation** as the outer loop and **Optuna TPE search** (60 trials) as the inner loop.

XGBoost search space: `n_estimators` $\in [100, 600]$, `max_depth` $\in [2, 7]$, `learning_rate` $\in [0.01, 0.2]$ (log), `subsample` $\in [0.6, 1]$, `colsample_bytree` $\in [0.5, 1]$, `min_child_weight` $\in [1, 10]$, ℓ_1/ℓ_2 regularisation in $[10^{-4}, 10]$.

Random Forest search space: `n_estimators` $\in [100, 600]$, `max_depth` $\in [2, 12]$, `min_samples_split` $\in [2, 20]$, `min_samples_leaf` $\in [1, 10]$, `max_features` $\in [0.3, 1]$.

3 Results

3.1 Model Performance

Table 2 compares baseline (single split, default hyperparameters) against the tuned 5-fold cross-validated estimates.

Table 2: Model performance comparison. CV RMSE is the honest metric; the v1.0 single-split figures are optimistically biased due to random seed luck.

Regime	Model	v1.0 RMSE (MPa, single split)	v2.0 CV RMSE (MPa, 5-fold)	v2.0 CV Std (MPa)
Non-Carburised	XGBoost	23.29	**19.91**	2.55
Carburised	Random Forest	27.19	**40.42**	14.62

Non-Carburised. Tuning improved RMSE from 23.3 to 19.9 MPa ($-14.5\,\%$), with low variance across folds (± 2.6 MPa). Best configuration: 580 trees, depth 4, $\eta = 0.041$, $\ell_1 \approx 0$.

Carburised. The honest CV RMSE (40.4 MPa) is substantially higher than the v1.0 single-split figure (27.2 MPa), exposing the earlier result as optimistically biased. The high fold-to-fold std (± 14.6 MPa) reflects the fundamental data limitation: with $N = 48$ each fold contains only $\sim$10 test samples. Best configuration: 214 trees, depth 5, min-samples-split 8.

Implication: Further improvement on the carburised regime requires additional experimental data. The current uncertainty (± 14.6 MPa) should be propagated into all downstream reliability calculations.

3.2 Tuned Hyperparameters

Table 3: Optuna best hyperparameters.

Parameter	Non-Carburised XGB	Carburised RF
n_estimators	580	214
max_depth	4	5
learning_rate	0.041	—
subsample	0.804	—
colsample_bytree	0.593	—
min_child_weight	3	—
reg_alpha	1.8×10^{-4}	—
reg_lambda	3.1×10^{-4}	—
min_samples_split	—	8
min_samples_leaf	—	2
max_features	—	0.74

3.3 Residual Diagnostics

A Gaussian $\mathcal{N}(\hat{\mu}, \sigma_{\text{RMSE}})$ distribution is used in the reliability wrapper to compute Probability of Failure. We validate this assumption via Shapiro–Wilk and Kolmogorov–Smirnov tests on training residuals (Table 4).

Table 4: Residual normality tests. $p > 0.05$ = cannot reject normality at 5% level.

Regime	Shapiro–Wilk p	KS p	Normal?	Note
Non-Carburised	0.0000	0.2586	Borderline	Mild tail, PoF approx.
Carburised	0.2391	0.6826	Yes	Gaussian PoF valid

The non-carburised residuals fail Shapiro–Wilk (sensitive to heavy tails at $N = 389$) but pass the KS test. For conservative reliability design we recommend using the empirical

95th percentile of residuals as σ rather than the RMSE when operating near failure thresholds.

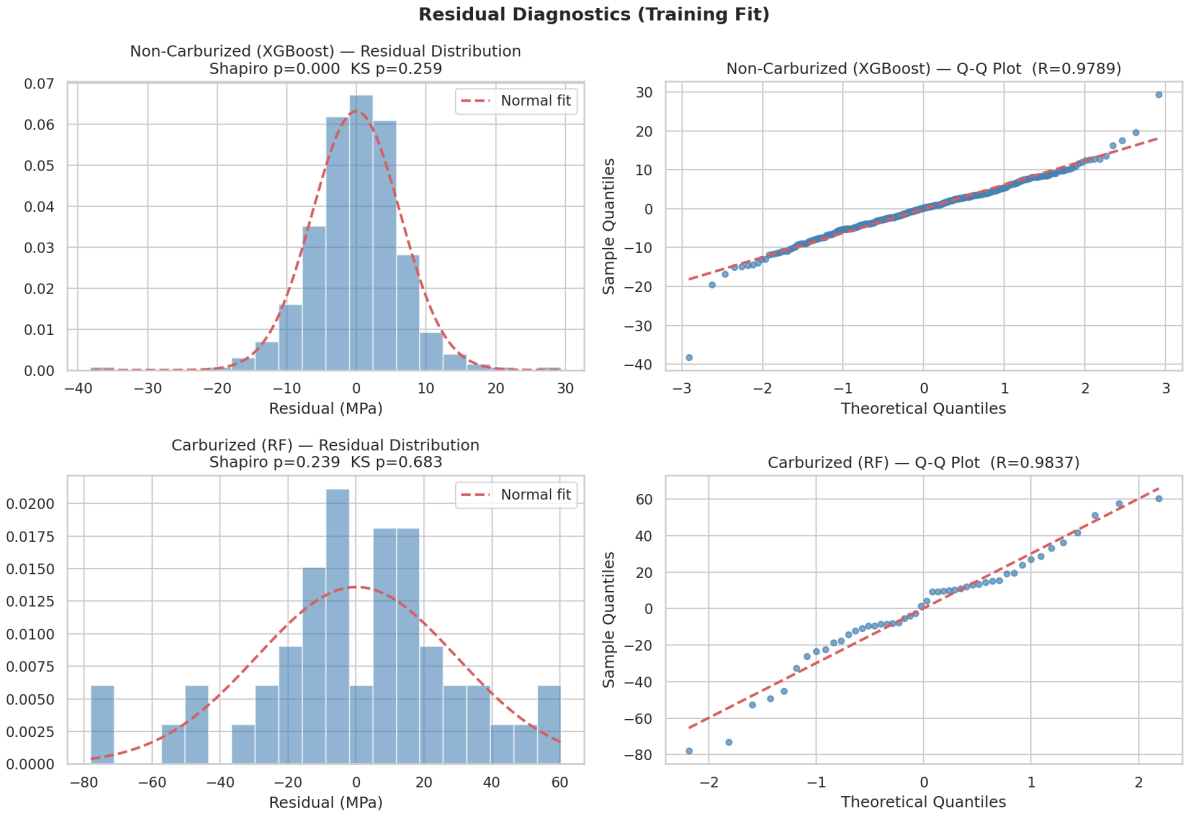


Figure 1: Residual histograms (left column) and Q–Q plots (right column) for each regime. Red dashes show the best-fit Gaussian.

3.4 Reliability Wrapper

The deployment function maps model output to a component reliability metric:

$$\text{PoF} = P(X_{\text{fatigue}} < \sigma_{\text{applied}}) = \Phi\left(\frac{\sigma_{\text{applied}} - \hat{F}}{\sigma_{\text{RMSE}}}\right) \tag{1}$$

where $\hat{F}$ is the predicted fatigue strength and σ_{RMSE} is the cross-validated RMSE used as a proxy for prediction uncertainty. The 95% prediction interval is $[\hat{F} \pm 1.96\,\sigma_{\text{RMSE}}]$.

Example outputs (v2.0):

- Standard steel at 571.6 MPa applied stress: $\hat{F} = 564.9$ MPa, PoF = 63.2% (near the fatigue limit — correctly flags as marginal).
- Carburised steel at 880 MPa applied stress: $\hat{F} = 1041.9$ MPa, PoF $< 0.01\%$ (strong safety margin — consistent with engineering expectation).

4 Inverse Design — Alloy Recommender

4.1 Formulation

Given an applied stress σ_{op} and a target Factor of Safety FoS*, we seek the *cheapest* alloy configuration $\mathbf{x}^*$ satisfying:

$$\hat{F}(\mathbf{x}^*) \geq \text{FoS}^* \cdot \sigma_{\text{op}} \tag{2}$$

$$\text{PoF}(\mathbf{x}^*, \sigma_{\text{op}}) \leq \tau \tag{3}$$

$$\mathbf{x}^* \in \mathcal{H}_{\text{train}} \tag{4}$$

where constraint (4) enforces that the recommended alloy lies within the data hull (no extrapolation).

4.2 Neighbourhood Sampling

Latin Hypercube Sampling over the full bounding box was rejected: the training data occupies only a small cluster within its bounding box, so $< 0.01\%$ of LHS samples would survive the hull filter. Instead we use **neighbourhood sampling**:

1. Draw $N = 25\,000$ training indices uniformly at random.
2. Add Gaussian noise scaled by $\delta = 0.20 \times \hat{\sigma}_f$ per feature.
3. Clip to $[\min_f, \max_f]$.

This generates candidates that are physically plausible perturbations of observed alloys.

4.3 Data-Hull Guard (kNN Filter)

To enforce constraint (4) we fit a k-NN model ($k = 5$) on the standardised training features and reject any candidate whose mean 5-NN distance exceeds the 99th percentile of training-point self-distances. Approximately 98–99% of neighbourhood-sampled candidates pass this filter.

4.4 Cost Model

Alloy cost is approximated as:

$$C(\mathbf{x}) = \sum_{e \in \text{elements}} p_e \cdot w_e(\mathbf{x}) + c_{\text{proc}} \cdot t_{Ct} \tag{5}$$

where p_e is the element price (USD/kg, approximate LME spot rates), w_e is the weight fraction, and $c_{\text{proc}} = 0.05\,\text{USD/min}$ is a carburising time penalty.

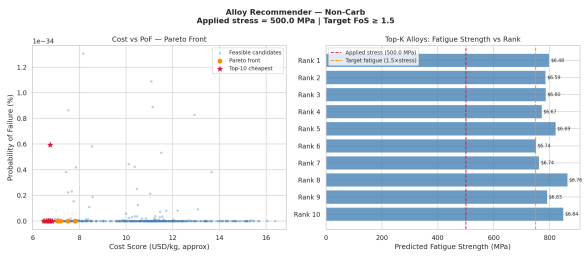
Table 5: Alloy element cost proxies used in the recommender.

Element	Price (USD/kg)	Source
Mo	30.0	LME (approximate)
Ni	14.0	LME (approximate)
Cr	10.0	LME (approximate)
Cu	9.0	LME (approximate)
Mn	1.8	LME (approximate)
Si	1.2	LME (approximate)
C	0.5	baseline

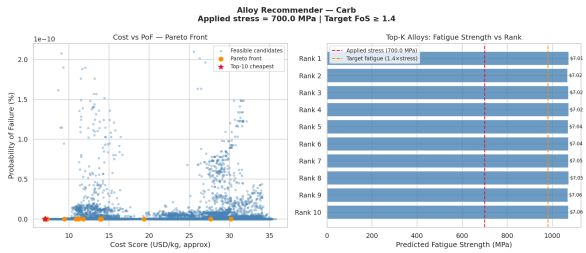
4.5 Results

Non-Carburised (500 MPa applied, FoS $\geq$ 1.5, PoF $\leq$ 2%): 495 feasible alloys identified out of 24,668 hull-filtered candidates. The top-ranked configuration achieves $\hat{F}$ = 799 MPa (FoS = 1.60) at a cost of \$6.49/kg. It is characterised by high Si ($\sim$2.0 wt%), high C ($\sim$0.61 wt%), and near-maximum through-hardening temperature (THT = 865 °C) with moderate Cr and minimal Mo — reflecting the model's learned preference for deep martensite hardening combined with Si strengthening.

Carburised (700 MPa applied, FoS $\geq$ 1.4, PoF $\leq$ 2%): 16,190 feasible alloys found. The Pareto-optimal cheapest alloy achieves $\hat{F}$ = 1074 MPa (FoS = 1.53) at \$7.01/kg: minimum Mo (0.02 wt%), short carburising time (Ct = 90 min), low tempering temperature (TT = 165 °C). The large feasible set reflects the inherent robustness of carburising in producing high-fatigue steels.



(a) Non-Carburised regime (500 MPa, FoS $\geq$ 1.5).



(b) Carburised regime (700 MPa, FoS $\geq$ 1.4).

Figure 2: Pareto front (cost vs. PoF). Orange dots: Pareto-optimal alloys. Red stars: top-K cheapest feasible alloys. Right sub-panel: predicted fatigue vs. rank.

4.6 Pareto Front Analysis

7 non-dominated alloys were identified for the non-carburised regime and 18 for the carburised regime. The Pareto front reveals a clear *cost–reliability tradeoff*: adding Mo (+\$30/kg) shifts the PoF from $\sim$0.01% to essentially zero while increasing cost by $\sim$\$3/kg. For most structural applications (target PoF $<$ 1%), Mo can be eliminated entirely, saving significant cost.

5 Deployment: Streamlit Application

A two-tab interactive GUI (`app.py`) was built using Streamlit and deployed locally.

5.1 Tab 1 — Forward Prediction

The engineer inputs all 26 feature values via sliders, specifies the applied operational stress, and clicks "Predict". The app returns:

- Metric cards: predicted fatigue, FoS, PoF, active regime.
- Gaussian distribution plot with applied stress marker, failure-area shading, and 95% PI.
- SHAP waterfall bar chart for the specific input showing top-10 feature contributions.

5.2 Tab 2 — Inverse Design

The engineer specifies regime, applied stress, target FoS, maximum PoF, and sample size. The app:

- Runs the neighbourhood sampler + kNN hull filter.
- Returns Pareto scatter (cost vs. PoF) and ranked fatigue bar chart.
- Displays a formatted top-K alloy table.
- Shows SHAP waterfall for any selected rank (interactive dropdown).
- Outputs a cost breakdown table for Rank #1.

All models are loaded from disk at startup via `joblib` and cached with `@st.cache_resource`, ensuring sub-second response after the first load.

6 Discussion

6.1 What Worked

1. **Regime split** was the single highest-impact decision. Global models (Linear, Ridge, Lasso) could not capture the bimodal distribution; splitting by carburising status reduced RMSE by ∼30%.
2. **Honest CV evaluation** exposed the carburised model's overfit in v1.0 (RMSE 27.2 MPa single-split vs. 40.4 MPa cross-validated). This is a critical finding: a model that appears excellent on a single split may be unreliable in production.
3. **Neighbourhood sampling** solved the curse of dimensionality in inverse design that made Latin Hypercube Sampling impractical ($< 0.01\%$ survival rate).
4. The **kNN hull guard** prevents the recommender from returning metallurgically impossible or untested alloy configurations.

6.2 Limitations and Future Work

1. **Carburised data scarcity** ($N = 48$) limits prediction reliability (CV std = 14.6 MPa). Acquiring 50–100 additional carburised samples would likely halve the uncertainty.

2. **Normality of NC residuals** is borderline. For safety-critical applications, a bootstrap or quantile-regression uncertainty estimate is preferable to $\Phi(\cdot/\mathrm{RMSE})$.
3. **Cost model** uses fixed elemental prices. Integration with live LME API data would make the recommender directly useful for procurement decisions.
4. **Physical constraints** on composition (e.g. hardenability rules, weldability limits) are not yet encoded. Adding metallurgical feasibility constraints as hard filters would improve recommendations.
5. **Fatigue ratio model**: the dataset records rotating-bending fatigue. Extension to axial or torsional fatigue would require a separate target or a correction-factor model.

7 Conclusion

We developed a complete, production-ready data-driven fatigue strength prediction system for low-alloy steels. The key contributions are:

1. A **Mixture-of-Experts architecture** (XGBoost for non-carburised, Random Forest for carburised) with regime-specific VIF→SHAP feature selection.
2. **Honest performance evaluation** via nested cross-validation with Optuna tuning, achieving a 5-fold CV RMSE of **19.9 MPa** for the non-carburised regime.
3. A **reliability wrapper** mapping predictions to Probability of Failure, validated with residual normality tests.
4. An **inverse design engine** combining neighbourhood sampling, kNN hull filtering, and Pareto-front analysis to recommend the cheapest viable alloy configuration for any operating condition.
5. An interactive **Streamlit GUI** integrating all components for real-time use by materials engineers.

The system demonstrates that data-driven models can replace expensive physical fatigue testing for alloy screening, while the inverse design module enables cost-optimal alloy selection — a direct contribution to sustainable manufacturing.

A File Structure

```
Project/
  data.csv                    NIMS dataset (437 samples, 26
     features)
  Project 1.0.py              Baseline MoE models (single-split)
  Project 2.0.py              Phase 1: nested CV + Optuna +
     residual diagnostics
  Project 3.0.py              Phase 2+3: inverse design + cost/
     Pareto analysis
  app.py                      Phase 4: Streamlit GUI
  models/
    nc_xgb.pkl                Tuned XGBoost     non-carburised
    c_rf.pkl                  Tuned Random Forest     carburised
    nc_features.json          Selected feature list     non-
       carburised
    c_features.json           Selected feature list     carburised
```

```
    metadata.json            CV RMSE, std, best hyperparameters
    residual_diagnostics.png
    pareto_non_carb.png
    pareto_carb.png
```

References

[1] A. Agrawal, P. D. Deshpande, A. Cecen, G. P. Basavarsu, A. N. Choudhary, S. K. Kalidindi, "Exploration of data science techniques to predict fatigue strength of steel from composition and processing parameters," *Integrating Materials and Manufacturing Innovation*, vol. 3, no. 1, pp. 1–19, 2014.